Voltage, current, resistance and Ohm's law

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The fundamental electrical quantities

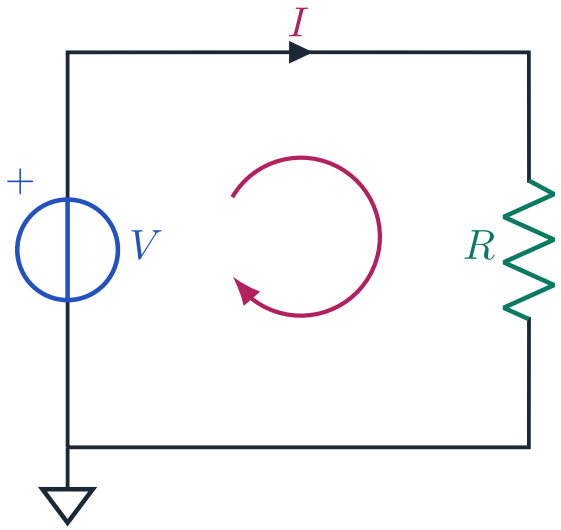
Every electrical circuit has three quantities working together: **voltage** pushes the charges, **current** is their movement, and **resistance** opposes that movement.

Quantity	Symbol	Unit	Instrument	How it is connected
voltage	*V*	volt (V)	voltmeter	in **parallel** with the component
current	*I*	ampere (A)	ammeter	in **series** with the component
resistance	*R*	ohm (Ω)	ohmmeter	on the component **disconnected** from the circuit

The electrical circuit

An **electrical circuit** is a closed path in which current can flow. It is made of:

- a **source**, which supplies the energy;
- a **load**, which uses the energy: here a resistance *R*;
- **conductors** (wires), which connect the elements;
- often a **switch**, which opens or closes the circuit.



If the circuit is **closed** current flows; if it is **open** (switch open or broken wire) the current is zero.

Voltage

Voltage (or **potential difference**) is the "push" that sets electric charges in motion. It is always measured **between two points** of the circuit.

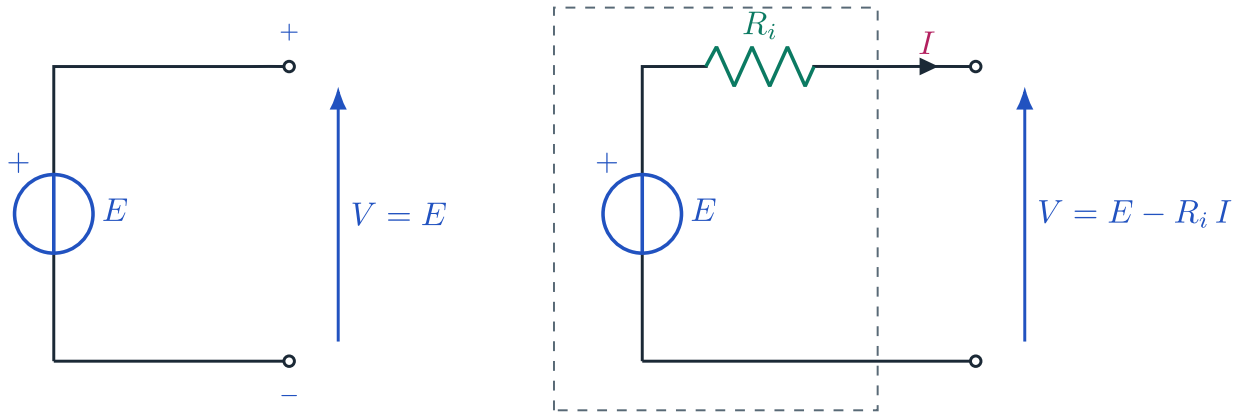
- Symbol: V · unit: **volt** (V).
- Multiples and submultiples: 1 kV = 1000 V · 1 mV = 0.001 V.
- It is marked with the **+** and **−** signs or with an arrow.

A useful comparison is with water: voltage is like the **difference in height** between two tanks; the bigger it is, the harder the water flows.

Typical values: AA cell 1.5 V · 9 V battery · USB port 5 V · car battery 12 V · household mains 230 V (alternating).

Voltage sources

A **voltage source** keeps a voltage between its two terminals, the **positive pole** (+) and the **negative pole** (−), by converting another form of energy into electrical energy: chemical (cell, battery), mechanical (alternator, dynamo), light (photovoltaic panel).



- **Ideal source:** the terminal voltage is always equal to E, whatever current is delivered.
- **Real source:** it has an **internal resistance** R_i. When it delivers current, part of the voltage is "lost" inside and less than E reaches the terminals.

> **Real source:** $V = E - R_i \cdot I$

E is called the **electromotive force** (e.m.f.): it is the voltage measured **with no load**, i.e. with the source disconnected ($I = 0$).

Example. A battery has $E = 9$ V and $R_i = 1\ \Omega$. If it delivers $I = 0.5$ A, the terminal voltage is $V = 9 - 1 \cdot 0.5 =$ **8.5 V**.

In the laboratory we use a **regulated power supply**: a source with adjustable voltage and current limiting, which behaves almost like an ideal source.

Electric current

Electric current is the **ordered movement of electric charges** in a conductor. In metals the moving charges are **electrons**.

Current intensity is the amount of charge flowing through a cross-section of the conductor in one second:

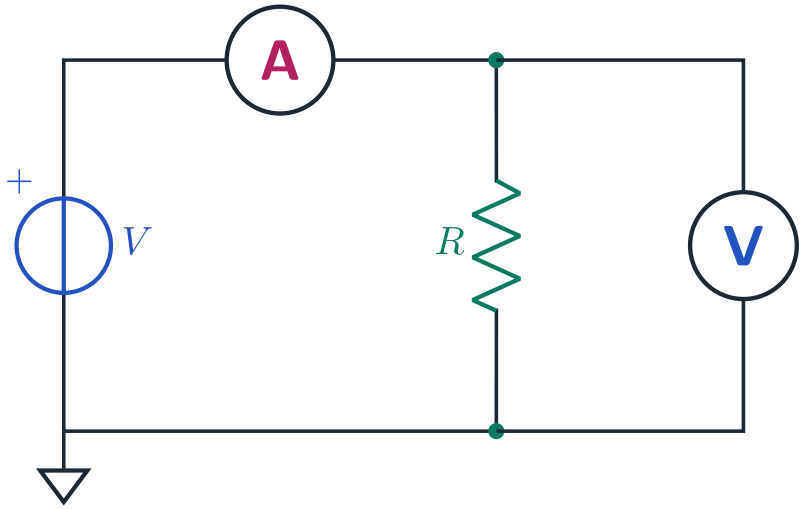
> **Current:** $I = Q / t$

- Q = electric charge, in **coulombs** (C); t = time, in seconds (s).
- Unit: **ampere** (A); 1 A = 1 C/s.
- Submultiples: 1 mA = 0.001 A · 1 µA = 0.000 001 A.
- **Conventional direction:** outside the source, current flows from the **+** pole to the **−** pole. Electrons actually move the opposite way.

Example. Q = 6 C flow through the wire in t = 3 s: I = 6 / 3 = **2 A**.

Typical values: LED 10–20 mA · car bulb about 2 A · car starter motor over 100 A.

Measuring voltage and current



- The **ammeter** is connected **in series**: the circuit must be opened so that the current flows through it. Its internal resistance is almost zero.
- The **voltmeter** is connected **in parallel**, across the component. Its internal resistance is very high.
- Careful: an ammeter connected in parallel creates a **short circuit** and may be damaged.

Resistance

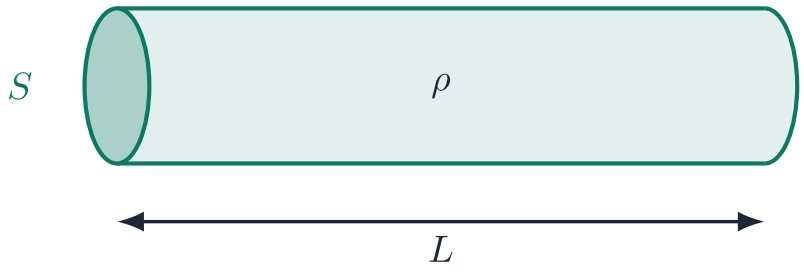
Electrical resistance is the **opposition** a material offers to the flow of current.

- Symbol: R · unit: **ohm** (Ω).
- Multiples: 1 kΩ = 1000 Ω · 1 MΩ = 1 000 000 Ω.

- **Conductors** (copper, aluminium, silver) have low resistance; **insulators** (plastic, glass, ceramics) very high.

The resistance of a wire depends on its **material**, **length** and **cross-section** (Ohm's second law):

> **Resistance of a wire:** $R = \rho \cdot L / S$



Symbol	Quantity	Unit
ρ (rho)	resistivity of the material	Ω · mm² / m
L	length of the wire	m
S	cross-section of the wire	mm²

The **longer** the wire, the **higher** the resistance; the **thicker** it is, the **lower** the resistance. Copper has $\rho \approx 0.0175$ Ω · mm²/m.

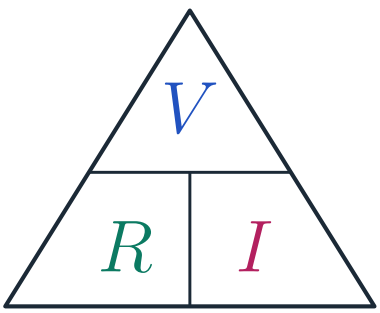
Example. A copper wire 100 m long with a 1.5 mm² cross-section has $R = 0.0175 \cdot 100 / 1.5 \approx$ **1.17 Ω**.

Resistors are components built to have a precise resistance value; the value is read from the **colour code** printed on the body.

Ohm's law

In a resistor the **current** is **directly proportional to the voltage** and **inversely proportional to the resistance**: if the voltage doubles the current doubles, if the resistance doubles the current halves.

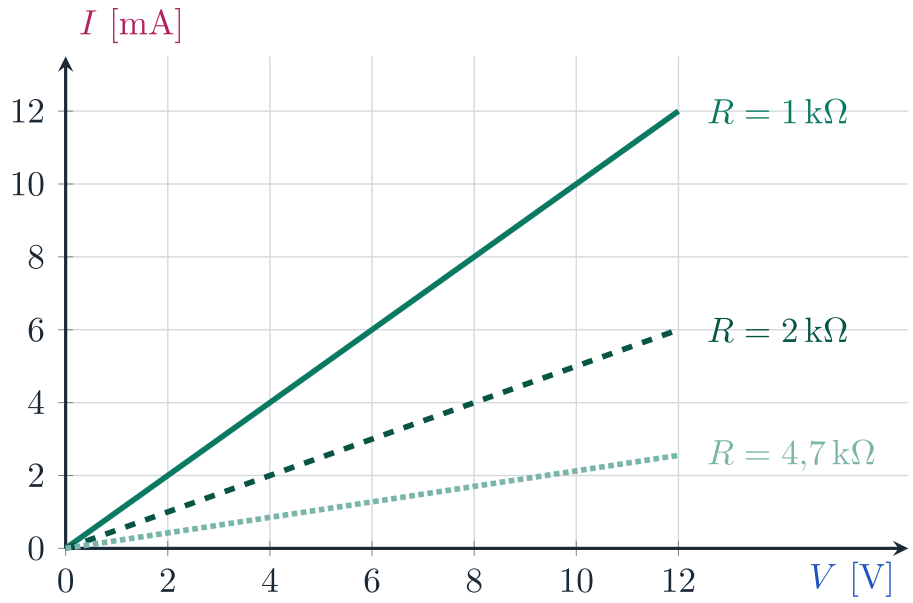
> **Current:** $I = V / R$
> **Voltage:** $V = R \cdot I$
> **Resistance:** $R = V / I$



How to use the triangle: cover the quantity to calculate with your finger and read the operation that remains. Covering V leaves $R \cdot I$; covering I leaves V / R; covering R leaves V / I.

The voltage–current characteristic

If you measure the current for several voltage values and plot the points, you get a **straight line through the origin**: the resistor is a **linear** component. The higher the resistance, the less steep the line.



Watch the units

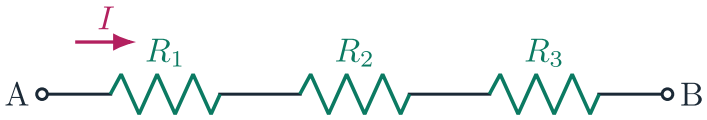
Formulas use V, A and Ω. In electronics the pair **kΩ and mA** is also handy: volts divided by kilohms gives milliamperes directly.

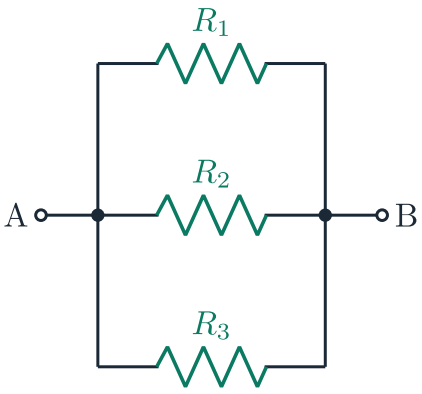
Voltage	Resistance	Current
V	Ω	A
V	kΩ	mA

Worked examples

- **Finding the current.** V = 12 V, R = 1 kΩ = 1000 Ω → I = 12 / 1000 = 0.012 A = **12 mA**.
- **Finding the voltage.** R = 220 Ω, I = 50 mA = 0.05 A → V = 220 · 0.05 = **11 V**.
- **Finding the resistance.** V = 9 V, I = 3 mA = 0.003 A → R = 9 / 0.003 = 3000 Ω = **3 kΩ**.
- **Using kΩ and mA.** V = 5 V, R = 2.2 kΩ → I = 5 / 2.2 ≈ **2.27 mA**.

Resistances in series and in parallel





Resistances in series

Resistances are **in series** when they are connected one after another, **with no branches**: the **same current** flows through all of them.

> **Equivalent resistance in series:** $R_{eq} = R_1 + R_2 + R_3 + \dots$

- The current is **the same** in all the resistances.
- The total voltage is the **sum** of the voltages: $V = V_1 + V_2 + V_3$.
- R_{eq} is always **larger** than the largest resistance.

Resistances in parallel

Resistances are **in parallel** when they are connected **between the same two nodes**: they have **the same voltage** across them.

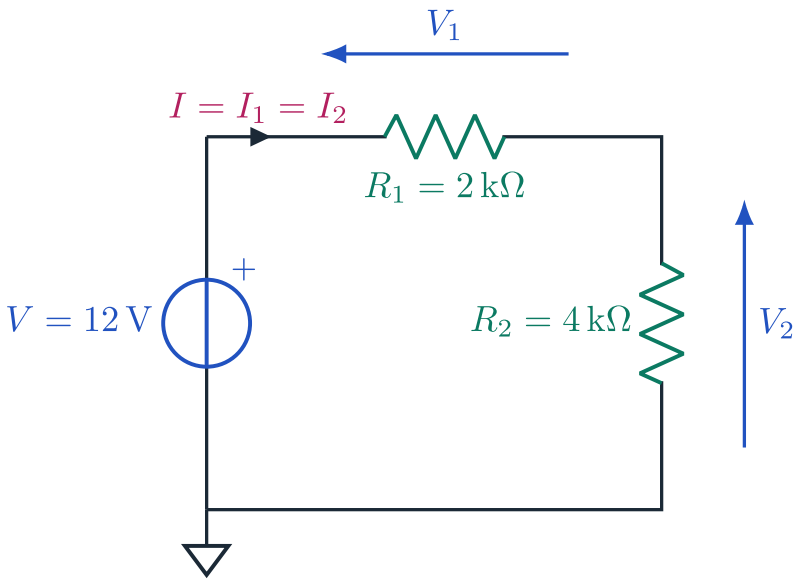
> **Equivalent resistance in parallel:** $1 / R_{eq} = 1 / R_1 + 1 / R_2 + 1 / R_3 + \dots$
>
> **Two resistances in parallel:** $R_{eq} = R_1 \cdot R_2 / (R_1 + R_2)$
>
> ***n* equal resistances in parallel:** $R_{eq} = R / n$

- The voltage is **the same** across all the resistances.
- The total current is the **sum** of the currents: $I = I_1 + I_2 + I_3$.
- R_{eq} is always **smaller** than the smallest resistance.

The symbol ∥ reads "in parallel with": $R_2 \parallel R_3$.

Example: two resistances in series with the source

V = 12 V, R_1 = 2 kΩ and R_2 = 4 kΩ in series.

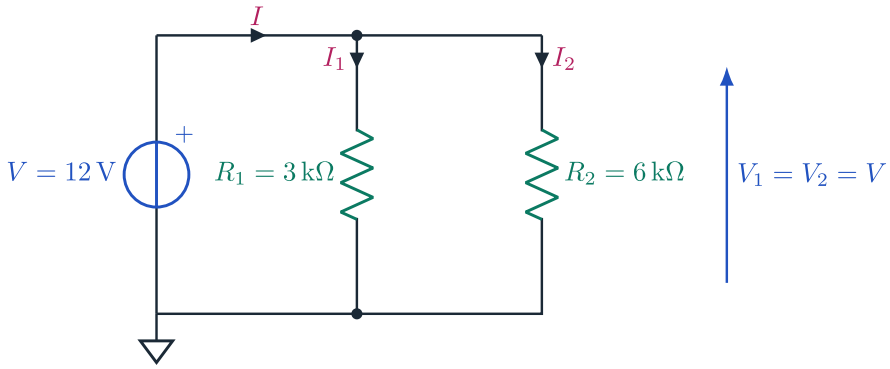


In series there is **only one path**: the current leaving the source flows first through R_1 and then through R_2, so **the current is the same** in both: $I = I_1 = I_2$. The voltages are **different** and add up.

- **equivalent resistance:** $R_{eq} = R_1 + R_2 = 2 + 4 =$ **6 kΩ**
- **current:** $I = V / R_{eq}$ = 12 V / 6 kΩ = **2 mA**
- **same current:** $I_1 = I_2 = I =$ **2 mA**
- **voltage across R_1:** $V_1 = R_1 \cdot I$ = 2 kΩ · 2 mA = **4 V**
- **voltage across R_2:** $V_2 = R_2 \cdot I$ = 4 kΩ · 2 mA = **8 V**
- **check:** $V_1 + V_2 = 4 + 8 = 12\ \mathrm{V} = V$ ✓ (the larger resistance takes the larger voltage)

Example: two resistances in parallel with the source

V = 12 V, R_1 = 3 kΩ and R_2 = 6 kΩ in parallel.



In parallel the two resistances are connected **to the same two nodes**, i.e. directly to the source terminals: **the voltage is the same** across both, $V_1 = V_2 = V$. The currents are **different** and add up.

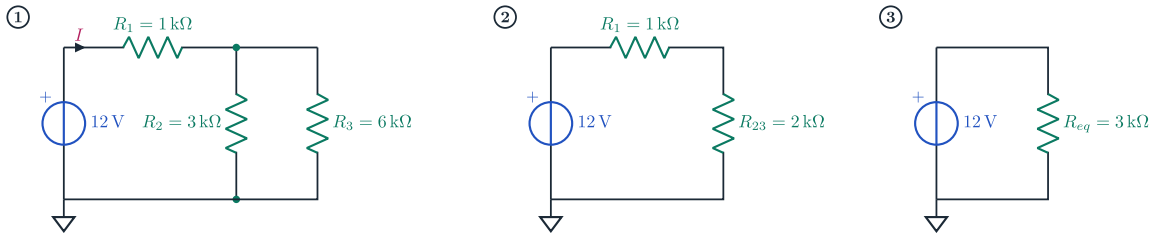
- **same voltage:** $V_1 = V_2 = V =$ **12 V**
- **current in R_1:** $I_1 = V / R_1 = 12\ \text{V} / 3\ \text{k}\Omega =$ **4 mA**
- **current in R_2:** $I_2 = V / R_2 = 12\ \text{V} / 6\ \text{k}\Omega =$ **2 mA**
- **total current:** $I = I_1 + I_2 = 4 + 2 =$ **6 mA** (the smaller resistance carries the larger current)
- **check with the equivalent resistance:** $R_{eq} = 3 \cdot 6 / (3 + 6) =$ **2 kΩ** → $I = 12\ \text{V} / 2\ \text{k}\Omega = 6\ \text{mA}$ ✓

How to solve a circuit with series and parallel resistances

- **1. Find the groups** of resistances that are clearly in series or in parallel, starting from the part of the circuit **furthest from the source**.
- **2. Replace** each group with its equivalent resistance and **redraw** the circuit.
- **3. Repeat** until a single resistance is left: the R_{eq} seen by the source.
- **4.** Calculate the **total current** with Ohm's law: $I = V / R_{eq}$.
- **5. Go back** through the redrawn circuits and calculate the voltage and current of each resistance with Ohm's law: in series the current is the same, in parallel the voltage is the same.
- **6. Check** the results: the voltages in series must add up to the total voltage, the currents in parallel to the total current.

Mixed example 1

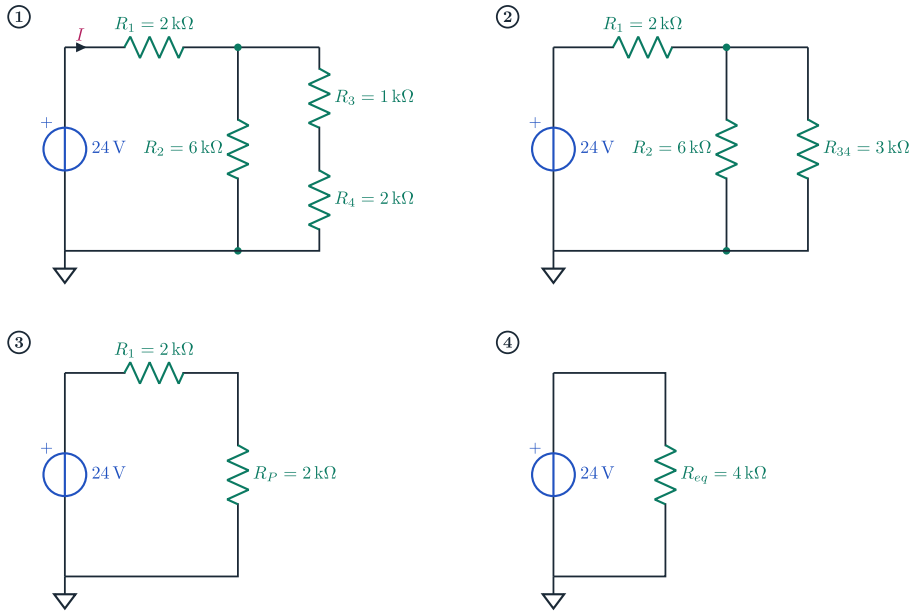
$V = 12$ V, $R_1 = 1$ kΩ in series with the parallel of $R_2 = 3$ kΩ and $R_3 = 6$ kΩ.



- **parallel:** $R_{23} = R_2 \parallel R_3 = 3 \cdot 6 / (3 + 6) = 18 / 9 =$ **2 kΩ**
- **series:** $R_{eq} = R_1 + R_{23} = 1 + 2 =$ **3 kΩ**
- **total current:** $I = V / R_{eq} = 12\ \text{V} / 3\ \text{k}\Omega =$ **4 mA** (the current through R_1)
- **voltage across R_1:** $V_1 = R_1 \cdot I = 1\ \text{k}\Omega \cdot 4\ \text{mA} =$ **4 V**
- **voltage across the parallel:** $V_{23} = R_{23} \cdot I = 2\ \text{k}\Omega \cdot 4\ \text{mA} =$ **8 V** (the same across R_2 and R_3)
- **currents in the parallel:** $I_2 = 8\ \text{V} / 3\ \text{k}\Omega \approx$ **2.67 mA** · $I_3 = 8\ \text{V} / 6\ \text{k}\Omega \approx$ **1.33 mA**
- **checks:** 4 + 8 = 12 V ✓ · 2.67 + 1.33 = 4 mA ✓

Mixed example 2

$V = 24$ V, $R_1 = 2$ kΩ in series with the parallel of $R_2 = 6$ kΩ and the series of $R_3 = 1$ kΩ and $R_4 = 2$ kΩ.



- **step 2, series:** $R_{34} = R_3 + R_4 = 1 + 2 =$ **3 kΩ**
- **step 3, parallel:** $R_\text{P} = R_2 \parallel R_{34} = 6 \cdot 3 / (6 + 3) = 18 / 9 =$ **2 kΩ**
- **step 4, series:** $R_\text{eq} = R_1 + R_\text{P} = 2 + 2 =$ **4 kΩ**
- **total current:** $I = 24$ V / 4 kΩ = **6 mA**
- **voltages:** $V_1 = 2$ kΩ · 6 mA = **12 V** · $V_\text{P} = 2$ kΩ · 6 mA = **12 V**
- **currents in the parallel:** $I_2 = 12$ V / 6 kΩ = **2 mA** · $I_{34} = 12$ V / 3 kΩ = **4 mA**
- **voltages across R_3 and R_4:** $V_3 = 1$ kΩ · 4 mA = **4 V** · $V_4 = 2$ kΩ · 4 mA = **8 V**
- **checks:** 12 + 12 = 24 V ✓ · 2 + 4 = 6 mA ✓ · 4 + 8 = 12 V ✓

Multiples and submultiples

Prefix	Symbol	Factor	Example
mega	M	$10^6 = 1\,000\,000$	1 MΩ = 1 000 000 Ω
kilo	k	$10^3 = 1000$	4.7 kΩ = 4700 Ω
milli	m	$10^{-3} = 0.001$	20 mA = 0.02 A
micro	μ	$10^{-6} = 0.000\,001$	500 μA = 0.0005 A

Formula summary

Ohm's law: $V = R \cdot I$; $I = V / R$; $R = V / I$

Current: $I = Q / t$

Resistance of a wire: $R = \rho \cdot L / S$

Real source: $V = E - R_{\text{i}} \cdot I$

Series: $R_{\text{eq}} = R_1 + R_2 + \ldots$

Parallel: $1 / R_{\text{eq}} = 1 / R_1 + 1 / R_2 + \ldots$; two resistances: $R_{\text{eq}} = R_1 \cdot R_2 / (R_1 + R_2)$